

WEEK 1 — BACKEND

Technical Design & Architecture Template

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TEAM	PROJECT	TRACK	WEEK
Error494:NotFound	OmniBus — Campus Transit & Safety Engine	Backend	1

Purpose of this document

This is the team's Week 1 backend design. It details system architecture, data models, business rules, APIs, security boundaries, parent email verification, and dual notification/email alert systems for OmniBus.

Judge-first principle

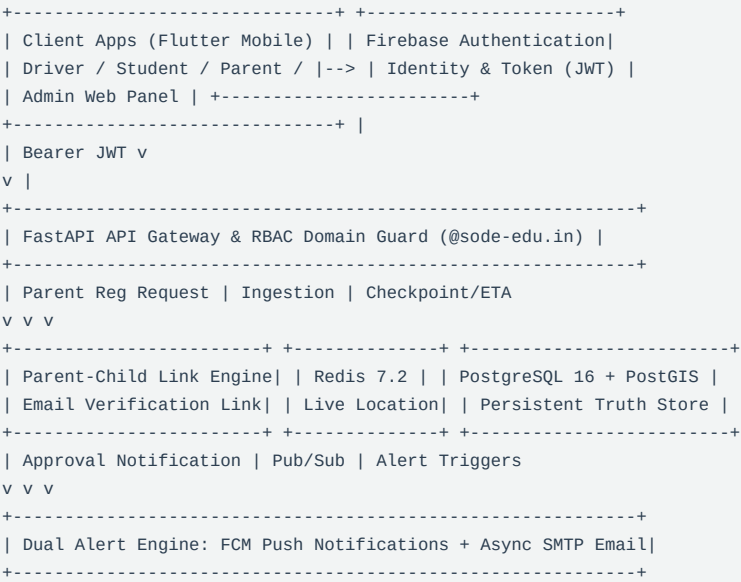
Prefer diagrams, tables, concise decisions, and concrete examples over generic prose. A judge should easily trace requirements from workflow → business rule → API → database.

1. System Overview

Item	Details
Backend Responsibility	Owns identity verification, RBAC authorization, parent-child email confirmation approval, real-time & batched GPS ingestion, Redis temporal caching, PostGIS geofence ETA, dual Push Notification & Email alerts, and 30s collision escalation.
Primary actors / roles	ADMIN: Fleet, route, & emergency management DRIVER: Trip lifecycle & GPS streaming STUDENT: Track assigned bus (@sode-edu.in domain); approves Parent link requests PARENT: Enters email, password & child email; linked after Student approval
Functional Decomposition	1. Auth & Student Domain Guard Module (@sode-edu.in check) 2. Parent-Child Registration & Email Confirmation Engine 3. Master Data & Active Trip Lifecycle CRUD 4. Real-time & Offline GPS Ingestion Engine 5. PostGIS Geofencing & Multi-Channel Alert Dispatcher (Push Notification + Email) 6. Multimodal Collision Escalation Handler
Main entities	User, Bus, Route, BoardingPoint, RouteStop, Trip, LocationPing, EmergencyEvent, Notification, ParentChildren, ParentLinkRequest

2. Backend Architecture

[ARCHITECTURE DIAGRAM]



Components

Component	Responsibility	Technology
API / Server	Handles REST APIs, WebSockets, Firebase JWT verification, RBAC, and Parent-Child approval flows.	Python 3.11 + FastAPI + Uvicorn
Database	Stores master data, spatial geometries, parent link requests, and location pings.	PostgreSQL 16 + PostGIS
Background worker	Flushes Redis pings to DB, triggers ETA calculations, dispatches FCM Push Notifications & Emails.	FastAPI Async Worker / ARQ Queue
Cache / queue	Caches live locations, pub/sub updates, notification idempotency locks (30m TTL).	Redis 7.2
External service	Provides user authentication JWTs, mobile push delivery, and transactional email dispatch.	Firebase Auth, FCM & SendGrid/SMTP Email

Key Architecture Decisions

ID	Decision	Reason	Trade-off / Limitation
AD-01	FastAPI Modular Monolith	High-throughput non-blocking WebSocket & REST API I/O.	Requires strict internal boundaries to avoid tight coupling.
AD-02	Parent Registration & Child Email Confirmation	Parent registers child email; link pending until student confirms via @sode-edu.in email link.	Parent cannot track bus until student explicitly approves link.
AD-03	Dual Notification Engine (Push + Email)	Dispatches push notifications via FCM and urgent alerts via SMTP/SendGrid email.	Slight latency addition for email worker dispatch.
AD-04	Drift SQLite Offline Queue	Zero data loss during mobile network blackouts in rural transit routes.	Sync engine must handle out-of-order timestamps and batch duplicate filtering.

3. Core Backend Workflows

Workflow 1: Parent Registration & Child Confirmation Flow

Parent Enters (Email, Password, Child Email) → Backend checks child exists? → If YES: Send verification request to child's @sode-edu.in email → Student approves parent request → Parent account approved & child bus linked ■.

Workflow 2: Dual Alert Engine (Notification & Email Dispatch)

Geofence / Delay / Collision Trigger → PostGIS Service → Redis Lock Check → Dispatch FCM Push Notification + Dispatch Async Email Alert (SendGrid/SMTP) to Parent/Student.

Workflow 3: Multimodal Accident Detection & 30s Escalation

Sensors Spike (G-Force > 3.5G + Deceleration) → 30s Driver Prompt → If unanswered/SOS → High Priority POST /api/v1/emergency/escalate → Admin Dashboard WebSockets + Urgency FCM Notification/Email.

4. Data Model

Database Schema definition for all core domain tables.

TABLE: users

Column	Type	PK / FK	Nullable	Default	Description
id	UUID	PK	No	gen_random_uuid()	Internal unique user ID
firebase_uid	VARCHAR(128)	—	No	—	External Firebase Auth UID
email	VARCHAR(255)	—	No	—	User email (@sode-edu.in for students)
full_name	VARCHAR(128)	—	No	—	Display full name
role	VARCHAR(20)	—	No	—	Role Enum (ADMIN/DRIVER/STUDENT/PARENT)
assigned_bus_id	UUID	FK	Yes	NULL	Ref to buses.id for driver/student

TABLE: parent_link_requests

Column	Type	PK / FK	Nullable	Default	Description
id	UUID	PK	No	gen_random_uuid()	Request identifier
parent_id	UUID	FK	No	—	Foreign key to users.id (Parent)
student_email	VARCHAR(255)	—	No	—	Target student @sode-edu.in email
token	VARCHAR(128)	—	No	—	Verification link token
status	VARCHAR(20)	—	No	'PENDING'	State (PENDING/APPROVED/REJECTED)

TABLE: trips

Column	Type	PK / FK	Nullable	Default	Description
id	UUID	PK	No	gen_random_uuid()	Trip unique ID
bus_id	UUID	FK	No	—	Assigned vehicle
driver_id	UUID	FK	No	—	Assigned driver
route_id	UUID	FK	No	—	Assigned route
status	VARCHAR(20)	—	No	'SCHEDULED'	State (SCHEDULED/ACTIVE/COMPLETED)

TABLE: location_pings

Column	Type	PK / FK	Nullable	Default	Description
id	UUID	PK	No	gen_random_uuid()	Ping entry ID
trip_id	UUID	FK	No	—	Foreign key to trips.id
location	GEOGRAPHY(POINT)	—	No	—	PostGIS Lat/Long coordinate
speed	FLOAT	—	No	0.0	Speed in km/h
recorded_at	TIMESTAMPTZ	—	No	—	Client recording timestamp

TABLE: notifications

Column	Type	PK / FK	Nullable	Default	Description
id	UUID	PK	No	gen_random_uuid()	Notification ID
user_id	UUID	FK	No	—	Recipient user reference
event_type	VARCHAR(30)	—	No	—	BUS_NEARBY / BUS_ARRIVED / DELAY
channel	VARCHAR(20)	—	No	'PUSH_AND_EMAIL'	Delivery channel (PUSH/EMAIL/BOTH)
created_at	TIMESTAMPTZ	—	No	now()	Dispatch timestamp

5. Business Logic & Rules

ID	Rule	Enforced By
BR-01	Student registration email MUST strictly end with @sode-edu.in domain.	FastAPI Domain Guard Middleware
BR-02	Parent account remains PENDING until child confirms link via @sode-edu.in email.	Parent Verification Engine
BR-03	All critical proximity, delay, & SOS events trigger dual Push Notification & Email.	Dual Notification / Email Dispatcher

BR-04	Drivers can only start, end, or stream pings for their assigned trips.	FastAPI RBAC Dependency Guard
BR-05	Duplicate notification/email alerts suppressed within 30-min window.	Redis Idempotency Key Lock Engine

6. API Design

API Summary

ID	Method	Endpoint	Auth	Role	Purpose
API-01	POST	/api/v1/auth/register-parent	Public	Parent	Registers parent & sends link confirmation to child's @sode-edu.in email
API-02	POST	/api/v1/parent/confirm-link	Required	Student	Student approves parent link request
API-03	POST	/api/v1/trips/start	Required	Driver	Starts assigned trip for driver
API-04	POST	/api/v1/gps/sync	Required	Driver	Batch sync real-time and offline buffered GPS pings
API-05	GET	/api/v1/trips/{id}/live	Required	Authorized	Get current live position & PostGIS ETA
API-06	POST	/api/v1/emergency/escalate	Required	Driver	High priority collision event / SOS escalation

Detailed Endpoint Contracts

API-01 — [POST /api/v1/auth/register-parent]

Field	Details
Purpose	Parent registration; verifies child exists and dispatches link request to child's @sode-edu.in email.
Authentication	Public / Optional Token
Validation	Valid parent email, password, and child email ending with @sode-edu.in
Success / Error	201 Created 400 Child Not Found / 409 Parent Already Exists

Request Payload:

```
{ "email": "parent@gmail.com", "password": "SecurePassword123!", "child_email": "child@sode-edu.in" }
```

Response Payload:

```
{ "success": true, "data": { "request_id": "req-9912", "status": "VERIFICATION_SENT_TO_CHILD", "message": "Confirmation link sent to child@sode-edu.in" } }
```

7. Authentication & Authorization

Authentication

Item	Implementation
Authentication mechanism	Firebase Authentication (Email/Password & Google OAuth)
Credential storage	Offloaded securely to Google Firebase Auth Infrastructure
Parent Link Approval	Child receives email link; student approval activates parent account link
Token / session lifetime	Firebase Short-lived JWT (1 hour lifetime)

RBAC Matrix

Action	Admin	Driver	Student	Parent
Create / Modify Master Data	✓ ALLOW	✗ DENY	✗ DENY	✗ DENY
Approve Parent Link Request	✓ ALLOW	✗ DENY	✓ ALLOW	✗ DENY
Start / End Assigned Trip	✗ DENY	✓ ALLOW	✗ DENY	✗ DENY
Post GPS Location Pings	✗ DENY	✓ ALLOW	✗ DENY	✗ DENY
View Assigned Bus & ETA	✓ ALLOW (All)	✓ ALLOW (Assigned)	✓ ALLOW (Assigned)	✓ ALLOW (Approved Child)

Security Controls

Control	Approach
Input validation	Pydantic v2 strict schema parsing & type validation on all REST endpoints
Authorization	FastAPI Dependency Guards verifying Firebase JWT, role scopes & approved parent links
Dual Alert Channel	Critical alerts dispatched via FCM Push Notification + SMTP/SendGrid Email
Rate limiting	Slowapi Redis-backed rate limiting (100 req/min REST, 10 pings/sec GPS)

8. Background Processing & Services

ID	Process / Service	Trigger	Purpose	Technology	Failure Handling
BG-01	Historical GPS Persister	Cron (Every 10s)	Flush Redis location pings to Postgres	FastAPI Async Worker	Retry 3x; log error on DB down
BG-02	Dual Notification / Email Dispatcher	GPS Geofence Event	Dispatch FCM Push Notification + SMTP Email	Redis Pub/Sub Worker	Suppress duplicate via 30m Redis lock

9. External Integrations

ID	Service	Purpose	Direction	Data Exchanged	Failure Behavior
EXT-01	Firebase Auth	Verify client JWT identity	FastAPI <- Firebase	ID Token, UID, Email	HTTP 401 Unauthenticated
EXT-02	FCM & SendGrid SMTP	Deliver Push Notifications & Email alerts	FastAPI -> FCM / SMTP	FCM Tokens, Email HTML, Alerts	Retry async queue; log failed emails

10. Error Handling & Reliability

Standard Error Response Format

```
{ "success": false, "error": { "code": "PARENT_LINK_PENDING_APPROVAL", "message": "Parent account linked to child@sode-edu.in is pending confirmation from student." } }
```

HTTP Status Conventions

Situation	Status	Example
Invalid input	400	Child email does not exist or invalid latitude/longitude format
Unauthenticated	401	Missing or expired Firebase JWT Bearer header
Unauthorized	403	Parent tracking child bus before student email approval
Not found	404	Requesting non-existent bus_id or trip_id
Conflict	409	Parent registration request for already linked child
Server failure	500	Unexpected database or SMTP email connection failure

Important Failure Scenarios

ID	Failure	Expected Backend Behavior
ERR-01	Mobile Network Outage during Trip	Driver app queues pings locally in Drift SQLite. On network restoration, /api/v1/gps/sync processes batch payload idempotently without missing points.
ERR-02	Student Email Link Expiration	Parent link verification token expires after 48 hours; parent can request resend via POST /api/v1/auth/resend-parent-link.
ERR-03	SMTP Email Server Timeout	Failed transactional emails are logged in notifications table with status PENDING and retried by background worker alongside FCM push.
ERR-04	Out-of-Order Batch GPS Upload	Ingestion filter compares ping timestamp with latest recorded trip timestamp; old out-of-order pings saved to history without moving live pointer backwards.

11. Testing Strategy

Test ID	Requirement / Flow	Scenario	Expected Result
T-01	REQ-01 Auth Domain	Student registers with student@gmail.com	403 Forbidden (INVALID_STUDENT_DOMAIN)
T-02	REQ-02 Parent Registration	Parent enters email, pass, & child@sode-edu.in	201 Created & Verification email sent to child
T-03	REQ-02 Child Confirmation	Student clicks email confirmation link	Parent status set to APPROVED ■
T-04	REQ-03 Dual Alerting	Geofence bus arrival trigger	FCM Push Notification + Async Email sent
T-05	REQ-04 Accident Alert	3.5G sensor spike without 30s confirmation	Escalated to Admin SOS Dashboard

Critical Test Coverage

- Parent self-registration & child email verification approval workflow
- Dual alert engine verification (FCM Push Notification + Async Email delivery)
- Student email domain restriction (@sode-edu.in validation)
- Role-based authorization & resource ownership restrictions
- Drift SQLite offline queue sync & PostGIS geofence trigger logic
- Multimodal collision calculation & 30s emergency confirmation countdown